
Wireless Network Design with DHCP

Course: The Bits and Bytes of Computer Networking

Date: 21/04/2026

Author: Florencia Horita

1. Introduction

This lab aims to design and implement a distributed wireless network using multiple Access Points in order to improve network coverage and performance.

A multi-floor environment is simulated where stable connectivity is required through proper Wi-Fi channel planning and the use of DHCP for automatic IP address assignment.

2. Network Topology

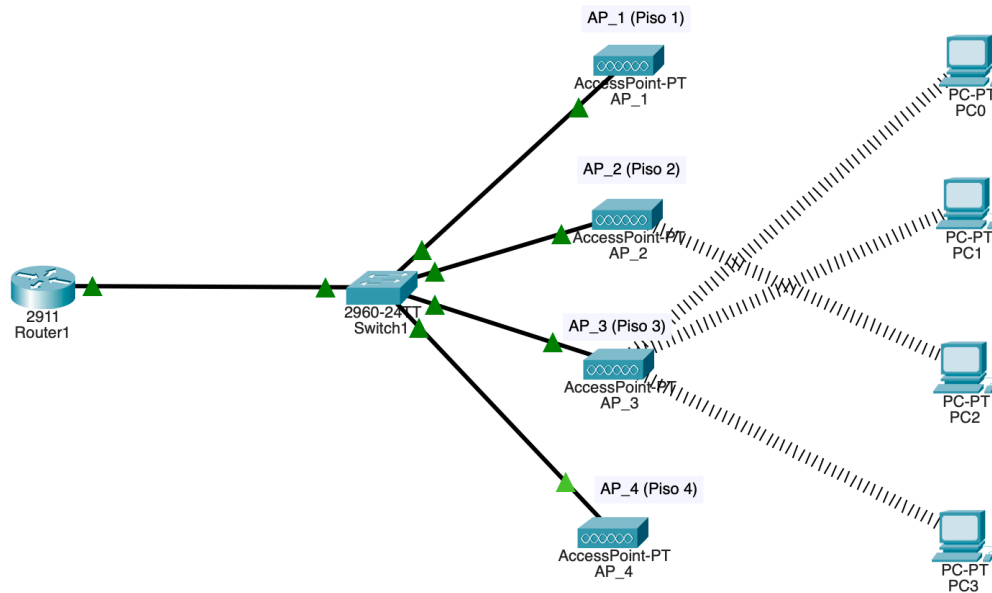
The network used in this lab is composed of:

- Four PCs (PC1, PC2, PC3, PC4)
- One switch (SW1)
- One router (R1)
- Four Access Points (AP1, AP2, AP3, AP4)

The connection is established as follows:

PCs → Access Points → Switch → Router

Figure 1 – Network Topology



3. Network Configuration

3.1 Access Point Configuration

Each Access Point was configured with the following parameters:

- SSID: Empresa_WIFI
- Security: WPA2-PSK
- Password: Empresa123

Channel distribution:

- AP1 → Channel 1
- AP2 → Channel 6
- AP3 → Channel 11
- AP4 → Channel 1

This configuration helps avoid interference in the 2.4 GHz band.

- AP1 (Floor 1) → Channel 1
- AP2 (Floor 2) → Channel 6

- AP3 (Floor 3) → Channel 11
- AP4 (Floor 4) → Channel 1

Figure 2 – Access Point Configuration

AP_1

PhysicalConfigAttributes

Port 1

Port Status

On

SSID

Empresa_WIFI

2.4 GHz Channel

1

Coverage Range (meters)

140,00

Authentication

Disabled

WPA-PSK

WEP

WPA2-PSK

WEP Key

PSK Pass Phrase

Empresa123

User ID

Password

Encryption Type

AES

3.2 Router Configuration

The router was configured as the network gateway with the following settings:

- IP: 192.168.1.1
- Subnet Mask: 255.255.255.0

Figure 3 – Router IP Configuration

```
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
.
```

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual  up          up
GigabitEthernet0/1  unassigned      YES unset   administratively down down
GigabitEthernet0/2  unassigned      YES unset   administratively down down
Vlan1          unassigned      YES unset   administratively down down
```

The router interface is in an “up/up” state, indicating it is active and operational.

3.3 DHCP Configuration

A DHCP pool was configured on the router to automatically assign IP addresses:

```
ip dhcp excluded-address 192.168.1.1
ip dhcp pool LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
```

This configuration allows devices to automatically obtain:

- IP address
- Subnet mask
- Gateway
- DNS server

Figure 4 – DHCP Configuration on the Router

```
ip dhcp excluded-address 192.168.1.1
!
ip dhcp pool LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
!
!
!
ip cef
no ipv6 cef
!
Router#
```

3.4 PC Configuration

The PCs were configured in automatic DHCP mode.

This allows them to receive network configuration without manual intervention.

Figure 5 – PC IP Configuration (ipconfig)

Global Settings

Display Name

Interfaces

Gateway/DNS IPv4

☒ DHCP
☐ Static

Default Gateway

DNS Server

Gateway/DNS IPv6

☒ Automatic
☐ Static

Default Gateway

DNS Server

4. Connectivity Tests

4.1 Connectivity Test

A connectivity test was performed from the PCs to the router:

```
ping 192.168.1.1
```

Result: Successful communication

Figure 6 – Successful Ping

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=55ms TTL=255
Reply from 192.168.1.1: bytes=32 time=42ms TTL=255
Reply from 192.168.1.1: bytes=32 time=40ms TTL=255
Reply from 192.168.1.1: bytes=32 time=38ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 55ms, Average = 43ms
```

5. Analysis

After the configuration, the communication flow works as follows:

- PCs connect to the nearest Access Point
- Access Points forward traffic to the switch
- The switch sends traffic to the router
- The router acts as gateway and DHCP server

The correct functioning of the following was validated:

- Wireless network (WLAN)
- Wi-Fi channel distribution
- Automatic IP assignment (DHCP)
- Device connectivity

6. Conclusion

This lab allowed the design of an efficient wireless network using multiple Access Points with a shared SSID and distributed channels to avoid interference.

The router's role as both gateway and DHCP server was highlighted, enabling automated device configuration and improving network scalability.

7. Simulation Considerations

It is important to note that Cisco Packet Tracer has certain limitations in wireless network simulation. In particular:

- Device selection of Access Points does not always reflect realistic behavior
- Roaming between Access Points is not accurately simulated
- Signal strength does not fully influence client connection behavior

Despite these limitations, the tool effectively helps understand the fundamental concepts of wireless network design.